# Laboratory Equipment and Safety

Hazards, glassware and reading scales

Name _____ Class _____ Date _____

**TYPICAL TOLERANCES (CLASS A)**

100 mL measuring cylinder ±0.5 mL · 10 mL pipette ±0.02 mL · 25 mL pipette ±0.03 mL

50 mL burette: each reading ±0.05 mL · 250 mL volumetric flask ±0.15 mL

Top-pan balance ±0.01 g · analytical balance ±0.0001 g

Percent uncertainty = uncertainty ÷ measured value × 100 %. Show units in every step.

## A. Hazards

**1.** State what each GHS pictogram means and give one example of a substance that carries it: (a) flame (b) flame over circle (c) corrosion (d) skull and crossbones (e) environment

**2.** (a) Which signal word shows the more serious hazard? (b) What is a safety data sheet, and when should you read it?

**3.** Identify **five** unsafe actions in this account and say what should be done instead: “Sam pushed her goggles up onto her head to see better, poured water into concentrated sulfuric acid, then heated ethanol in a test tube over a Bunsen flame, pointing the tube at her partner. She sipped her drink while waiting and poured the leftover copper(II) sulfate solution down the sink.”

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## B. Choosing equipment

**4.** Choose the best equipment for each task: (a) adding exactly 25.00 mL of a solution to a flask (b) roughly 50 mL of water to dissolve a solid (c) making exactly 250.0 mL of a solution (d) adding acid drop by drop while measuring the volume (e) weighing 0.5000 g of a solid

**5.** Calculate the percent uncertainty of: (a) 0.50 g on a top-pan balance and 0.5000 g on an analytical balance (b) 10.0 mL measured with a 100 mL measuring cylinder and 10.00 mL delivered by a 10 mL pipette. What do your answers show?

## C. Burette readings

**6.** A student records these titrations. Rough: 0.00 mL → 24.10 mL. Run 1: 1.20 mL → 24.70 mL. Run 2: 0.40 mL → 23.85 mL. Run 3: 0.05 mL → 23.55 mL. (a) Calculate each titre. (b) Calculate the mean and standard deviation of runs 1–3 and report the result.

**7.** Explain why you read the bottom of the meniscus with your eye level with it, and name the error caused by reading from above.

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## D. Challenge

**8.** Each burette reading is uncertain by ±0.05 mL, and a titre is the difference of two readings. (a) Calculate the percent uncertainty of a titre of 23.50 mL. (b) Suggest one way to reduce the percent uncertainty of the titre.

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